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In recent years, several studies have been carried out on topological descriptors as they find extensive applications in the development of QSAR and QSPR studies [1–12]. Consequently, there is renewed attention for the advancement and development of new topological descriptors which will further extend the potentiality of QSAR/QSPR approaches and their applications [13–16]. QSAR/QSPR models have been employed in a number of areas to characterize molecular structures, biological networks, and information science to correlate structures or networks with their properties for which the degree and distance measures from graphs have a significant impact. In the context of drug discovery guest-host interactions play critical roles, and thus rapid computations of their properties could result in significant cost saving. These techniques largely depend on quantitative measures, some of which are based on graph distance degree measures that are commonly employed to analyse structural differences of molecular compounds [13,17]. These types of modelling are critical to environmental modelling of pollutants and petroleum products as in such emissions typically a mixture of large number of polycyclic aromatic compounds are found. For example, in a major accidental fire involving rubber tires numerous polycyclic aromatic compounds were found in the emission [18]. Consequently, mathematical modelling techniques are highly desirable for rapid assessment of the potential hazards to the environment so that adequate remediation measures can be taken.

Topological indices and their applicability to circumcised donut benzenoid systems, kekulenes and related drugs have also been considered [19]. The graph of a molecule is denoted by G in which all the atoms grouped in a set $V(G)$, called the vertex set, and the bonds denoted by a set $E(G)$, the set of edges of G. For a vertex u, the degree d_u is the number of edges incident to u. Two types of vertex-degree based Zagreb indices have been well studied due to their diverse applications, and an analogous of the first type Zagreb index is called the forgotten index [20]. The first Zagreb index computes the square of vertex degrees by taking the sum over all vertices, whereas the forgotten index computes the sum of cubes of vertex degrees over all vertices. The additive combination of the forgotten and the second Zagreb indices in the ratio of 1: 2 result in a hyper-Zagreb index and it has received considerable attention [21,22] over the years. In the series, we now take the ratio of 1: 1, and propose two types of new indices called the bi-Zagreb and tri-Zagreb indices, which are stated below:

$$BM(G) = \sum_{u \in V(G)} d_u^2 + \sum_{uv \in E(G)} d_u d_v = \sum_{uv \in E(G)} (d_u + d_v + d_u d_v)$$

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desirable for rapid assessment of the potential hazards to the environment so that adequate remediation measures can be taken.

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$$TM(G) = \sum_{u \in V(G)} d_u^3 + \sum_{uv \in E(G)} d_u d_v = \sum_{uv \in E(G)} (d_u^2 + d_v^2 + d_u d_v)$$

In another perspective, the combination of two topological indices in fractional ratio was introduced in [23] and here we introduce ten new types of topological indices by taking various combinations of geometric, harmonic, bi-Zagreb and tri-Zagreb indices and demonstrate their efficacy.

$$GH(G) = \sum_{uv \in E(G)} \frac{\sqrt{d_u d_v}(d_u + d_v)}{2} \qquad \text{(Geometric} - \text{Harmonic)}$$

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$$GBM(G) = \sum_{uv \in E(G)} \frac{\sqrt{d_u d_v}}{d_u + d_v + d_u d_v} \text{ (Geometric – Bi Zagreb)}$$

$$GTM(G) = \sum_{uv \in E(G)} \frac{\sqrt{d_u d_v}}{d_u^2 + d_v^2 + d_u d_v} \text{ (Geometric – Tri Zagreb)}$$

$$HG(G) = \sum_{uv \in E(G)} \frac{2}{\sqrt{d_u d_v}(d_u + d_v)} \text{ (Harmonic – Geometric)}$$

$$HBM(G) = \sum_{uv \in E(G)} \frac{2}{(d_u + d_v + d_u d_v)(d_u + d_v)} \text{ (Harmonic – Bi Zagreb)}$$

$$HTM(G) = \sum_{uv \in E(G)} \frac{2}{\left(d_u^2 + d_v^2 + d_u d_v\right)(d_u + d_v)} \text{ (Harmonic – Tri Zagreb)}$$

$$BMG(G) = \sum_{uv \in E(G)} \frac{d_u + d_v + d_u d_v}{\sqrt{d_u d_v}} \text{ (Bi Zagreb – Geometric)}$$

$$BMH(G) = \sum_{uv \in E(G)} \frac{(d_u + d_v + d_u d_v)(d_u + d_v)}{2} \text{ (Bi Zagreb – Harmonic)}$$

$$TMG(G) = \sum_{uv \in E(G)} \frac{d_u^2 + d_v^2 + d_u d_v}{\sqrt{d_u d_v}} \text{ (Tri Zagreb – Geometric)}$$

$$TMH(G) = \sum_{uv \in E(G)} \frac{(d_u^2 + d_v^2 + d_u d_v)(d_u + d_v)}{2} \text{ (Tri Zagreb – Harmonic)}$$

We study the effectiveness of the proposed topological descriptors compared to the existing degree and distance types of topological descriptors using a data set consisting of 55 benzenoid hydrocarbons and present the associated linear regression equations for several physicochemical properties along with their implication in predicting physicochemical properties of hydrocarbons.

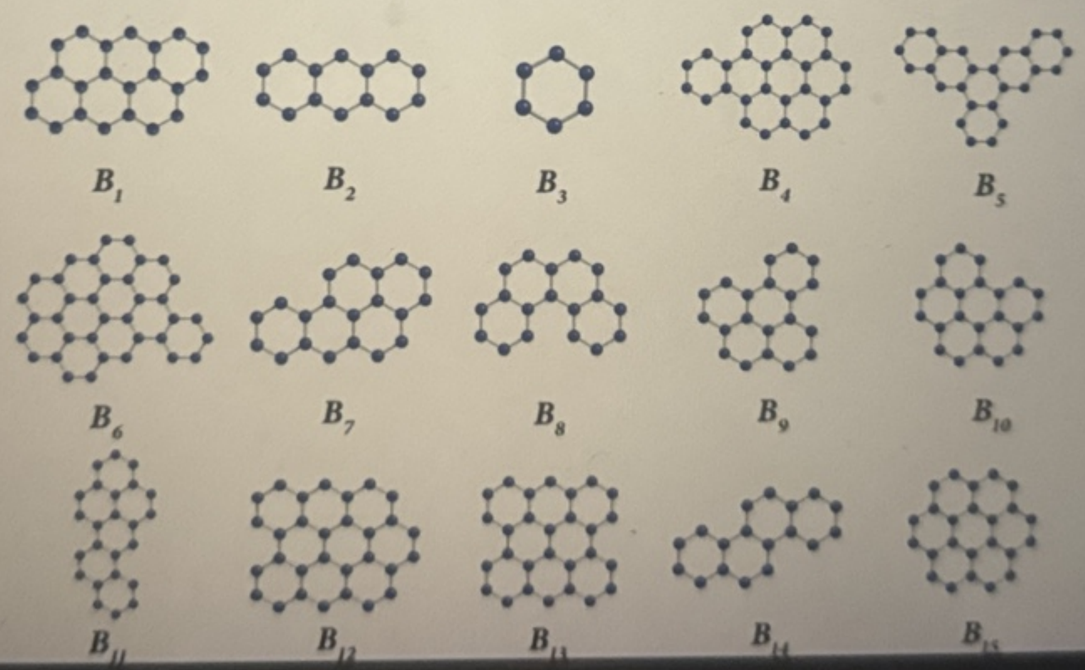
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erties of hydrocarbons.

Computation of hybrid geometric-harmonic-Zagreb descriptors for benzenoid hydrocarbons

In this section, we compute the topological descriptors derived from taking the ratio of geometric, harmonic, bi-Zagreb and tri-Zagreb indices by considering a data set of 55 benzenoid hydrocarbons as shown in Figures 1 and 2. For this purpose, we partition the bonds of hydrocarbons using the notation given below:

$$d_{kj} = |\{uv \in E(G) : d_u = k, d_v = j\}|$$



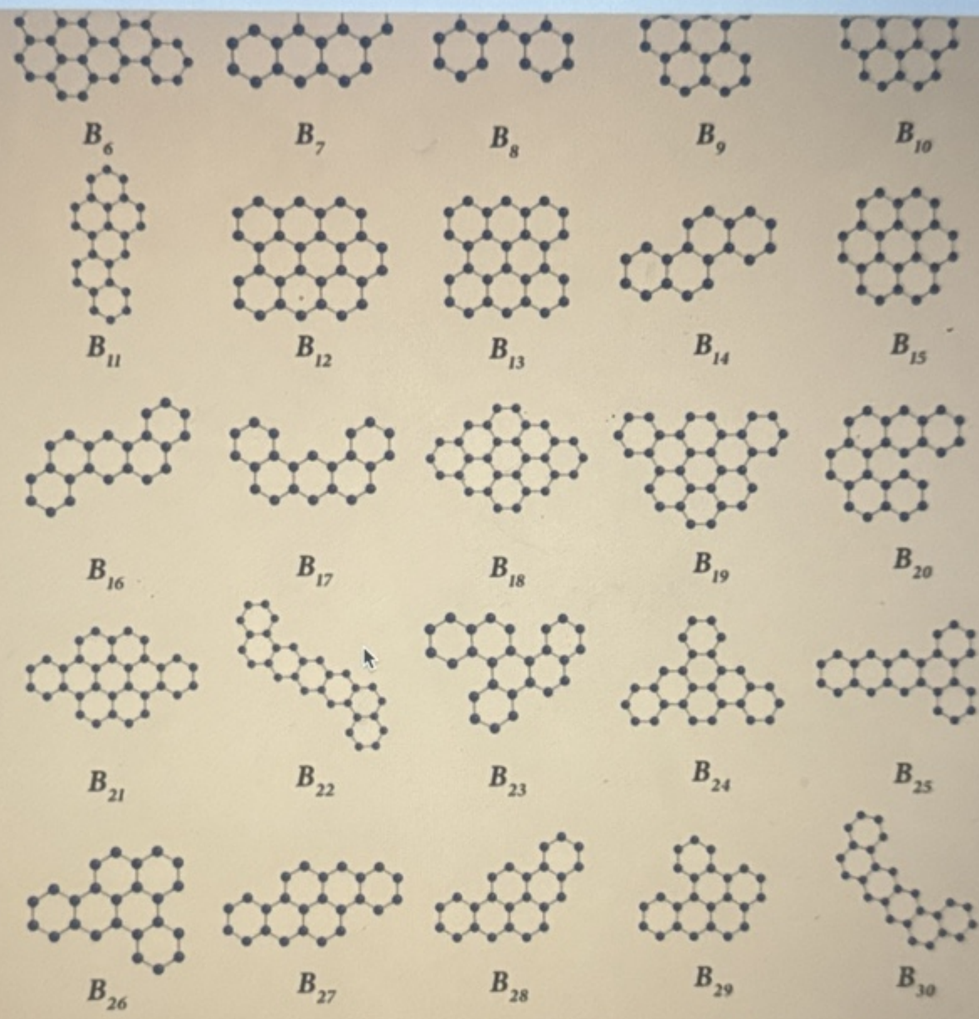


Figure 1. Benzenoid hydrocarbons: B_1-Anthanthrene, B_2-Anthracene, B_3-Benzene, B_4-Benzo[a]coronene, B_5-Benzo[h]pentaphene, B_6-Benzo[a]ovalene, B_7-Benzo[a]pyrene, B_8-Benzo[c]phenanthrene, B_9-Benzo[e]pyrene, B_{10}-Benzo[ghi]perylene, B_{11}-Benzo[pqr]picene, B_{12}-Benzobisanthene, B_{13}-Bisanthene, B_{14}-Chrysene, B_{15}-Coronene, B_{16}-Dibenz[a,h]anthracene, B_{17}-Dibenz[a,j]anthracene, $B18$-Dibenz[Bc,Kl]Coronene, $B19$-Dibenzo[a,g]coronene, B_{20}-Dibenzo[a,ghi]perylene, B_{21}-Dibenzo[a,j]coronene, B_{22}-Dibenzo[a,l]pentacene, B_{23}-Dibenzo[c,p]chrysene, B_{24}-Dibenzo[h,rst]Pentaphene, B_{25}-Dibenzo[a,c]tetracene, B_{26}-Dibenzo[a,e]pyrene, B_{27}-Dibenzo[a,h]pyrene, B_{28}-Dibenzo[a,i]pyrene, B_{29}-Dibenzo[a,l]pyrene, B_{30}-Dibenzo[a,l]tetracene (B_1,B_2,B_7-B_{10},B_{14}-B_{17} and B_{26}-B_{29} are reproduced from [11]).

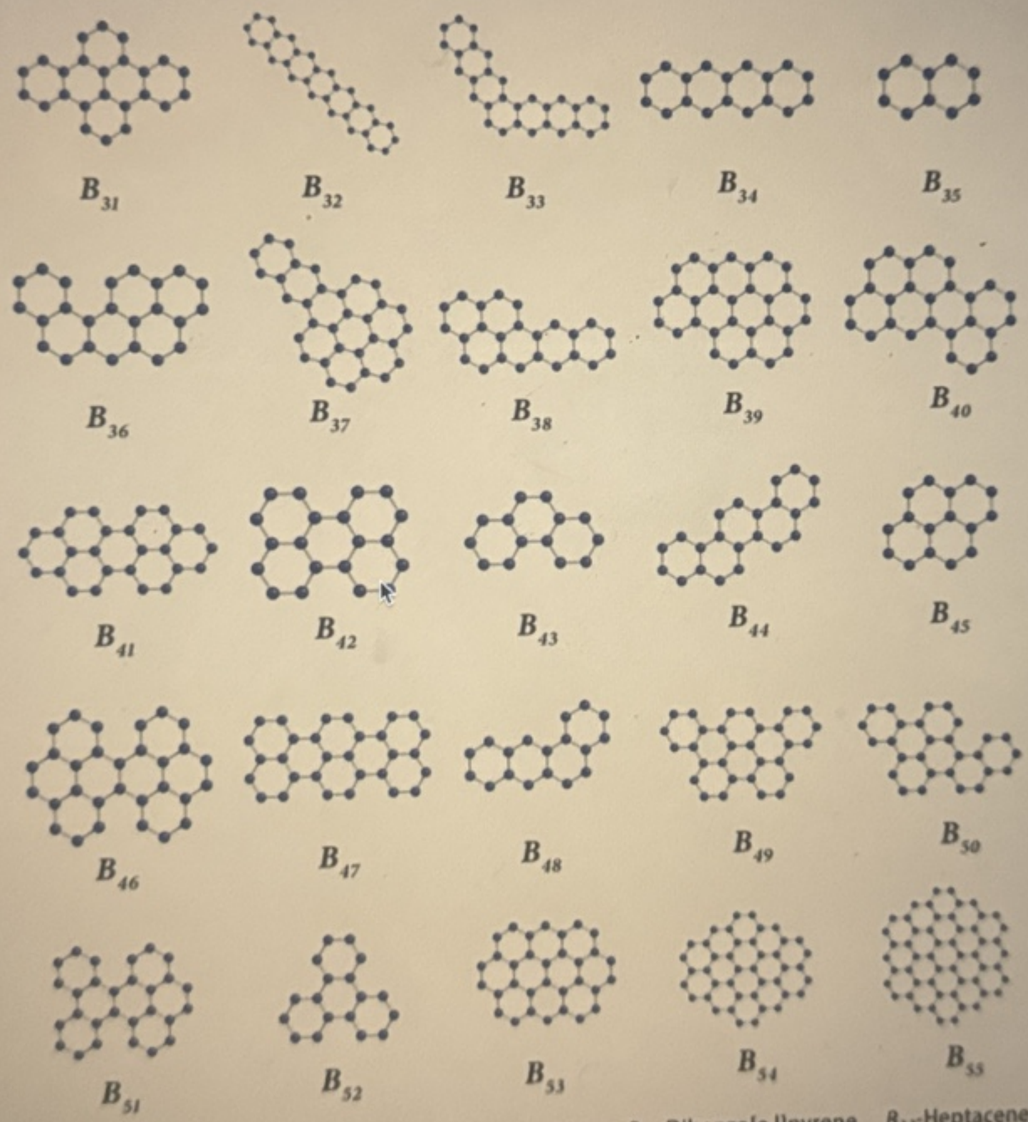


Figure 2. Benzenoid hydrocarbons continuation: B_{31}-Dibenzo[e,l]pyrene, B_{32}-Heptacene, B_{33}-Heptaphene, B_{34}-Naphthacene, B_{35}-Naphthalene, B_{36}-Naphtho[1,2-a]pyrene, B_{37}-Naphtho[2,3-a]coronene, B_{38}-Naphtho[2,3-a]pyrene, B_{39}-Naphtho[5,4,3-abc]coronene, B_{40}-Naphtho[8,1,2-bcd]perylene, B_{41}-Peropyrene, B_{42}-Perylene, B_{43}-Phenanthrene, B_{44}-Picene, B_{45}-Pyrene, B_{46}-Pyreno[4,5-e]pyrene, B_{47}-Terylene, B_{48}-Tetraphene, B_{49}-Tribenzo[b,n,pqr]perylene, B_{50}-Tribenzo[a,fg,op]tetracene,

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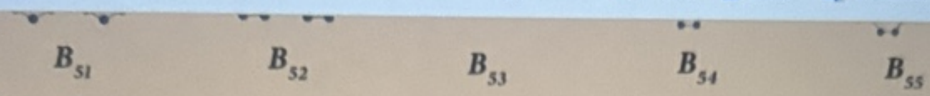


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Graph polynomials are analytical expressions derived from a graph to build various algebraic models to determine graph information [24,25]. The *M*-polynomial is one such expression introduced to quantify the degree-based topological indices of chemical

structures using calculus operators [26]. A primary advantage of the *M*-polynomial technique is that several topological indices can be computed simultaneously from the expression. The *M*-polynomials have been employed extensively in deriving QSAR expressions for toxicity and other properties of antiepileptic and β-dione compounds [27,28].

The *M*-polynomial of a graph *G* is formally defined as follows:

$$M(G;x,y) = \sum_{k\le j} d_{k,j} x^k y^j$$

The benzenoid hydrocarbons shown in Figures 1, 2 and 3 constitute the data set for the current study and they have partition classes $d_{2,2}$, $d_{2,3}$ and $d_{3,3}$ as given in Table 1, thus the above-described *M*-polynomial is simplified as follows:

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$$M(G;x,y)=d_{2,2}x^2y^2+d_{2,3}x^2y^3+d_{3,3}x^3y^3$$

The *M*-polynomials of B_i, $1\le i\le 55$, are presented in Table 1, the numerical quantities for bi-Zagreb, tri-Zagreb, geometric-harmonic, geometric-bi-Zagreb, geometric-tri-Zagreb, harmonic-geometric, harmonic-bi-Zagreb, harmonic-tri-Zagreb, bi-Zagreb-geometric, bi-Zagreb-harmonic, tri-Zagreb-geometric and tri-Zagreb-harmonic descriptors are obtained by constructing suitable differential and integral operators [29,30] with the aid of the MATLAB interface because the manual procedure is tedious to compute the descriptor values which are presented in Table 2. For instance, we show the steps to obtain the *BMG* index for anthracene compound by defining its suitable operator as $D_x^{\frac{1}{2}}S_y^{\frac{1}{2}}+D_y^{\frac{1}{2}}S_x^{\frac{1}{2}}+D_x^{\frac{1}{2}}D_y^{\frac{1}{2}}|_{x=y=1}$, where

$$D_x^{\frac{1}{2}}(f(x,y))=\sqrt{x\frac{\partial f(x,y)}{\partial x}}\sqrt{f(x,y)},$$

$$D_y^{\frac{1}{2}}(f(x,y))=\sqrt{y\frac{\partial f(x,y)}{\partial y}}\sqrt{f(x,y)},$$

$$S_x^{\frac{1}{2}}(f(x,y))=\sqrt{\int_0^x\frac{f(t,y)}{t}dt}\sqrt{f(x,y)},$$

$$S_y^{\frac{1}{2}}(f(x,y))=\sqrt{\int_0^y\frac{f(x,t)}{t}dt}\sqrt{f(x,y)}.$$

From Table 1, we have $M(B_2;x,y)=f(x,y)=6x^2y^2+8x^2y^3+2x^3y^3$ and the operators can be applied as follows:

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$$\sqrt{J_0 \quad t}$$

$$S_y^{\frac{1}{2}}(f(x,y)) = \sqrt{\int_0^y \frac{f(x,t)}{t}\,dt}\sqrt{f(x,y)}.$$

From Table 1, we have $M(B_2;x,y) = f(x,y) = 6x^2y^2 + 8x^2y^3 + 2x^3y^3$ and the operators can be applied as follows:

$$S_y^{\frac{1}{2}} f(x,y) = \frac{6x^2y^2}{\sqrt{2}} + \frac{8x^2y^3}{\sqrt{3}} + \frac{2x^3y^3}{\sqrt{3}}$$

$$D_x^{\frac{1}{2}} S_y^{\frac{1}{2}} f(x,y) = \frac{6\sqrt{2}x^2y^2}{\sqrt{2}} + \frac{8\sqrt{2}x^2y^3}{\sqrt{3}} + \frac{2\sqrt{3}x^3y^3}{\sqrt{3}}$$

Similarly, we have

Table 1. Bond partitions of 55 benzenoid hydrocarbons and their *M*-polynomials.

Compounds	*Atoms*	*Bonds*	$d_{2,2}$	$d_{2,3}$	$d_{3,3}$	$M(G;x,y)$
Anthanthrene	22	27	6	12	9	$6x^2y^2 + 12x^2y^3 + 9x^3y^3$
Anthracene	14	16	6	8	2	$6x^2y^2 + 8x^2y^3 + 2x^3y^3$
Benzene	6	6	6	0	0	$6x^2y^2$
Benzo[a]coronene	28	35	8	12	15	$8x^2y^2 + 12x^2y^3 + 15x^3y^3$
Benzo[h]pentaphene	26	31	9	14	8	$9x^2y^2 + 14x^2y^3 + 8x^3y^3$
Benzo[a]ovalene	36	46	8	16	22	$8x^2y^2 + 16x^2y^3 + 22x^3y^3$
Benzo[a]pyrene	20	24	7	10	7	$7x^2y^2 + 10x^2y^3 + 7x^3y^3$
Benzo[c]phenanthrene	18	21	8	8	5	$8x^2y^2 + 8x^2y^3 + 5x^3y^3$
Benzo[e]pyrene	20	24	8	8	8	$8x^2y^2 + 8x^2y^3 + 8x^3y^3$
Benzo[ghi]perylene	22	27	7	10	10	$7x^2y^2 + 10x^2y^3 + 10x^3y^3$
Benzo[pqr]picene	24	29	8	12	9	$8x^2y^2 + 12x^2y^3 + 9x^3y^3$
Benzobisanthene	30	38	7	14	17	$7x^2y^2 + 14x^2y^3 + 17x^3y^3$
Bisanthene	28	35	8	12	15	$8x^2y^2 + 12x^2y^3 + 15x^3y^3$
Chrysene	18	21	8	8	5	$8x^2y^2 + 8x^2y^3 + 5x^3y^3$
Coronene	24	30	6	12	12	$6x^2y^2 + 12x^2y^3 + 12x^3y^3$
Dibenz[a,h]anthracene	22	26	8	12	6	$6x^2y^2 + 12x^2y^3 + 6x^3y^3$
Dibenz[a,j]anthracene	22	26	8	12	6	$8x^2y^2 + 12x^2y^3 + 6x^3y^3$

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Anthracene	14	16	6	8	2	$6x^2y^2 + 8x^2y^3 + 2x^3y^3$
Benzene	6	6	6	0	0	$6x^2y^2$
Benzo[a]coronene	28	35	8	12	15	$8x^2y^2 + 12x^2y^3 + 15x^3y^3$
Benzo[a]ovalene	26	31	9	14	8	$9x^2y^2 + 14x^2y^3 + 8x^3y^3$
Benzo[a]pyrene	36	46	8	16	22	$8x^2y^2 + 16x^2y^3 + 22x^3y^3$
Benzo[c]phenanthrene	20	24	7	10	7	$7x^2y^2 + 10x^2y^3 + 7x^3y^3$
Benzo[e]pyrene	18	21	8	8	5	$8x^2y^2 + 8x^2y^3 + 5x^3y^3$
Benzo[ghi]perylene	20	24	8	8	8	$8x^2y^2 + 8x^2y^3 + 8x^3y^3$
Benzo[pqr]picene	22	27	7	10	10	$7x^2y^2 + 10x^2y^3 + 10x^3y^3$
Benzobisanthene	24	29	8	12	9	$8x^2y^2 + 12x^2y^3 + 9x^3y^3$
Bisanthene	30	38	7	14	17	$7x^2y^2 + 14x^2y^3 + 17x^3y^3$
Chrysene	28	35	8	12	15	$8x^2y^2 + 12x^2y^3 + 15x^3y^3$
Coronene	18	21	8	8	5	$8x^2y^2 + 8x^2y^3 + 5x^3y^3$
Dibenz[a,h]anthracene	24	30	6	12	12	$6x^2y^2 + 12x^2y^3 + 12x^3y^3$
Dibenz[a,j]anthracene	22	26	8	12	6	$6x^2y^2 + 12x^2y^3 + 6x^3y^3$
Dibenz[Bc,Kl]Coronene	22	26	8	12	6	$8x^2y^2 + 12x^2y^3 + 6x^3y^3$
Dibenzo[a,g]coronene	30	38	6	16	16	$6x^2y^2 + 16x^2y^3 + 16x^3y^3$
Dibenzo[a,ghi]perylene	32	40	10	12	18	$10x^2y^2 + 12x^2y^3 + 18x^3y^3$
Dibenzo[a,j]coronene	26	32	7	13	12	$7x^2y^2 + 13x^2y^3 + 12x^3y^3$
Dibenzo[a,l]pentacene	32	40	10	12	18	$10x^2y^2 + 12x^2y^3 + 18x^3y^3$
Dibenzo[c,p]chrysene	30	36	8	20	8	$8x^2y^2 + 20x^2y^3 + 8x^3y^3$
Dibenzo[h,rst]Pentaphene	26	31	11	11	9	$11x^2y^2 + 11x^2y^3 + 9x^3y^3$
Dibenzo[a,c]tetracene	28	34	10	12	12	$10x^2y^2 + 12x^2y^3 + 12x^3y^3$
Dibenzo[a,e]pyrene	26	31	9	14	8	$9x^2y^2 + 14x^2y^3 + 8x^3y^3$
Dibenzo[a,h]pyrene	24	29	9	10	10	$9x^2y^2 + 10x^2y^3 + 10x^3y^3$
Dibenzo[a,i]pyrene	24	29	8	12	9	$8x^2y^2 + 12x^2y^3 + 9x^3y^3$
Dibenzo[a,l]pyrene	24	29	8	12	9	$8x^2y^2 + 12x^2y^3 + 9x^3y^3$
Dibenzo[a,l]tetracene	24	29	9	10	10	$9x^2y^2 + 10x^2y^3 + 10x^3y^3$
Dibenzo[e,l]pyrene	26	31	8	16	7	$8x^2y^2 + 16x^2y^3 + 7x^3y^3$
Heptacene	24	29	10	8	11	$10x^2y^2 + 8x^2y^3 + 11x^3y^3$
Heptaphene	30	36	6	24	6	$6x^2y^2 + 24x^2y^3 + 6x^3y^3$
Naphthacene	30	36	7	22	7	$7x^2y^2 + 22x^2y^3 + 7x^3y^3$
Naphthalene	18	21	6	12	3	$6x^2y^2 + 12x^2y^3 + 3x^3y^3$
naphtho[1,2-a]pyrene	10	13	6	4	1	$6x^2y^2 + 4x^2y^3 + 1x^3y^3$
Naphtho[2,3-a]coronene	24	29	8	12	9	$8x^2y^2 + 12x^2y^3 + 9x^3y^3$
Naphtho[2,3-a]pyrene	32	40	8	16	16	$8x^2y^2 + 16x^2y^3 + 16x^3y^3$
Naphtho[3,4,3-abc]coronene	24	29	7	14	8	$7x^2y^2 + 14x^2y^3 + 8x^3y^3$
Naphtho[8,1,2-bcd]perylene	30	38	7	14	17	$7x^2y^2 + 14x^2y^3 + 17x^3y^3$
Pentapyrene	26	32	8	12	12	$8x^2y^2 + 12x^2y^3 + 12x^3y^3$
Perylene	20	32	8	12	12	$8x^2y^2 + 12x^2y^3 + 12x^3y^3$
Phenanthrene	14	24	8	8	6	$8x^2y^2 + 8x^2y^3 + 6x^3y^3$
Picene	22	30	7	10	7	$7x^2y^2 + 10x^2y^3 + 7x^3y^3$
Pyrene	16	19	6	8	5	$6x^2y^2 + 8x^2y^3 + 5x^3y^3$
Terylene	30	37	10	12	15	$10x^2y^2 + 12x^2y^3 + 15x^3y^3$

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